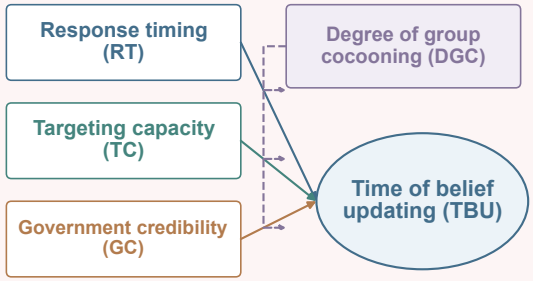
I Model design framework

Freeze case and population

One allegation + official response
Synthetic profiles; fixed network

Tested associations and moderation



Moderation: DGC × RT, TC and GC

54-cell controlled factorial design

Response timing (RT)		
Low	Medium	High
2	4	7

Targeting capacity (TC)	
Low	High
.20	.80

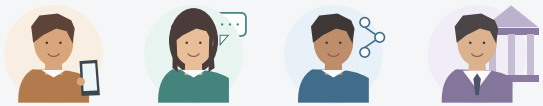
Government credibility (GC)		
Low	Medium	High
.25	.50	.75

Degree of group cocooning (DGC)		
Low	Medium	High
0	.20	.40

300 LLM users · 54 scenarios × 10 rounds

One stochastic trajectory per parameter cell

II Agent design and simulation



Rumor blogger	Inside rumor cocoon users	Outside rumor cocoon users	Government account
SOURCE	n = 48	n = 252	50 recipients

Demographics · MBTI · trust · habits
Prior score · 3-step memory · actual exposure

Begin round t

Retain the previous 3 rounds
Publish the official response if t = RT

Construct the visible feed

Following, reposts and revisits
≤30 earlier comments per visible root

LLM-Agent decision

Integrate all 11 assigned conditions
Score, comment and repost choices

Record decisions and update states

Store b and r; TBU = r − b (r ≥ RT)
Social actions become visible at t + 1

t ← t + 1

Publishers are scripted; audience users call the LLM.

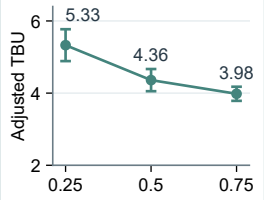
III Results and inference

First-belief TBU = r − b
HC3 OLS + F / t tests + Holm

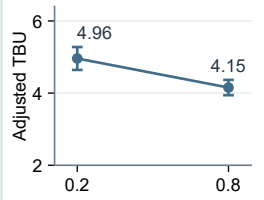
54 scenario means · 757 observed updates

DGC level difference: adjusted p = .272

Government credibility (GC): p < .001

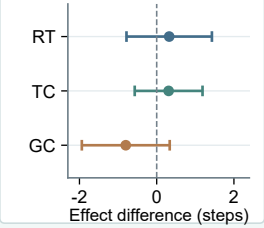


Targeting capacity (TC): p < .001



Cocooning interactions

All adjusted p > .81



Response timing (RT)

Adjusted p < .001

